

Jacob Horne

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EDUCATION

University of California, Irvine

June 2028

Bachelor of Science in Computer Science; Campuswide Honors Collegium

GPA: 3.92

Coursework: Data Structures & Algorithms, Operating Systems, Distributed Systems, System Design, ML & Data Mining
Artificial Intelligence, Databases, Computer Architecture, Software Engineering, Computer Vision, Probability & Statistics

EXPERIENCE

Sandia National Laboratories

Aug 2026 – Present

Software Engineer Intern

Livermore, CA

- Building full-stack applications and backend services that make Sandia AI research accessible through user-facing tools

Capital One

Jun 2026 – Aug 2026

Software Engineer Intern

New York, NY

- Built Databricks sampler for offline replay used by analysts and ML teams to test recommender models and campaigns; sped replay **28×** across **~800M daily outcomes** and broader date ranges with **18 PySpark checks** and Airflow automation
- Migrated production routing for **95% ML-ranked and 5% random-control users** from a network API to an in-process SDK, preventing timeout-driven misrouting and cutting latency **25% for 60M+ monthly users**
- Validated routing and promotional rankings with unit and E2E tests using AWS CloudWatch, ECS, Fargate, and S3

UCI Digital Learning Lab

Dec 2025 – Present

Research Assistant

Irvine, CA

- Evaluated Advocate/Skeptic reasoning on ASAP dataset essays using LLM-as-judge rubrics across four writing traits
- Analyzed token log-probs with NumPy/Pandas/Matplotlib to relate confidence features to rubric-based reasoning quality
- Co-authored **3 publications** on LLM confidence and multi-agent reasoning at **ICLR Workshop**, **ACL SRW**, **GEM** (2026)
- Investigating token-level confidence across multi-turn text-to-SQL query generation, execution, and iterative repair cycles

California Institute for Telecommunications and Information Technology

Apr 2026 – Present

Machine Learning Research Assistant

Irvine, CA

- Developed PyTorch/GRU models replacing **hours-long** heat-transfer FEA simulations for wall-temperature prediction
- Extended a GRU into a physics-informed neural network constrained by heat-transfer equations and initial conditions
- Achieved **12,000×** **faster inference than heat-transfer simulations** at scale with **$R^2 = 0.99$** and **MAE = 1.54°C**

Commit the Change at UCI

Oct 2025 – Jun 2026

Full-Stack Developer

Irvine, CA

- Built React/TypeScript scheduling/quota platform with quota creation and staff access, supporting **8,000+ patients/year**
- Developed Node.js/Express.js/PostgreSQL provider directory for **24+ providers**, saving **12+ staff hrs/week**

SENS Psychology

Sep 2025 – Jan 2026

Software Development Intern

Arlington, VA

- Built Next.js/TypeScript intake and ticketing platform used by **30+ staff across 1,200+ monthly patient interactions**
- Cut **patient intake time 50% and manual emails 75%** via Node.js/PostgreSQL automation and Azure notifications

PROJECTS

Agonus - AI Agent Trading Platform | Blockchain at UCI

Oct 2025 – Apr 2026

- Architected LangChain ReAct agents (GPT-4o-mini) with tools for market data, trade execution, planning, and memory
- Designed Celery/Redis scheduling, locking, and crash recovery to reliably orchestrate **10+ concurrent autonomous agents**
- Contributed to decentralized AI trading platform with on-chain tournaments, live leaderboards, and token rewards

Prop.Intel - AI Property Risk Platform | IrvineHacks 2026

Feb 2026 – Mar 2026

- Built AI property-risk platform using ATTOM/FEMA data to generate real-time disaster-risk reports for U.S. properties
- Developed ridge regression model across **3,100+ counties and 8 hazard types** to quantify nationwide disaster exposure
- Built interactive property-risk dashboards for ownership timelines, hazard breakdowns, and AI-generated risk summaries

TECHNICAL SKILLS

Languages: Python, C++, SQL, Java, TypeScript, JavaScript, C, R

Backend/Web: React, Next.js, Node.js, Express.js, FastAPI, REST APIs, PostgreSQL, Redis, Celery

Data/ML: PyTorch, scikit-learn, PySpark, Databricks, Airflow, TensorFlow, NumPy, Pandas, Matplotlib, LangChain, LangGraph

Systems/Cloud: Linux, Bash, Docker, Git, CI/CD, CUDA, AWS, ECS, Fargate, S3, CloudWatch, Azure